

Prime Number Checker Function

and

Function to Find and Display Primes in a Range

This function will check if a given number is prime. A prime number is a number greater than 1 that has no positive divisors other than 1 and itself. We'll optimize the check by testing divisibility up to the square root of the number.

This function will iterate through a given range and use the isPrime function to identify and collect prime numbers. It will then display these numbers in a user-friendly format.

```
/**
 * Check if a number is prime.
 *
 * A prime number is a natural number greater than 1 that is
not a product
 * of two smaller natural numbers. This function efficiently
checks for
 * primality by testing divisibility up to the square root of the
number.
 *
 * @param int $number The number to check.
 * @return bool True if the number is prime, false otherwise.
 */
function isPrime(int $number): bool
{
    if ($number <= 1) {
        return false; // 0 and 1 are not prime numbers
    }
    if ($number == 2) {
        return true; // 2 is the only even prime number
    }
    if ($number % 2 == 0) {
        return false; // Eliminate other even numbers
    }

    // Check divisibility from 3 up to the square root of the
number
    for ($i = 3; $i <= sqrt($number); $i += 2) {
        if ($number % $i == 0) {
            return false;
        }
    }
    return true;
}
```

Detailed Explanation of the Code

Prime Number Checker Function (isPrime)

Function Purpose: Determines if a number is prime.

Parameters:

\$number: The number to check for primality.

Return Value:

Returns true if the number is prime, false otherwise.

Logic:

Initial Checks:

Numbers less than or equal to 1 are not prime.
2 is the only even prime number.
Other even numbers greater than 2 are not prime.

Return Value:

None (void). It directly outputs the results.

Logic:

Range Validation: Ensures that the start of the range is not greater than the end.

Prime Collection:

Iterates through each number in the range.
Uses the isPrime function to check if a number is prime.
Collects all prime numbers in an array.

Result Display:

If no primes are found, outputs an appropriate message.
If primes are found, displays them in a comma-separated list.

```
/**
 * Find and display all prime numbers in a given range.
 *
 * This function iterates through each number in the
specified range, checks
 * if it is prime using the isPrime() function, and collects all
prime numbers
 * in an array. It then displays these prime numbers.
 *
 * @param int $start The start of the range.
 * @param int $end The end of the range.
 * @return void
 */
function displayPrimesInRange(int $start, int $end): void
{
    if ($start > $end) {
        echo "Invalid range: Start should be less than or equal to
End.";
        return;
    }

    $primes = [];
    for ($i = $start; $i <= $end; $i++) {
        if (isPrime($i)) {
            $primes[] = $i;
        }
    }

    if (empty($primes)) {
        echo "No prime numbers found in the range $start to
$end.";
    } else {
        echo "Prime numbers in the range $start to $end: " .
implode(', ', $primes);
    }
}

// Example usage with error handling and clear output
try {
    $start = 10;
    $end = 50;
    displayPrimesInRange($start, $end);
} catch (Exception $e) {
    echo "Error: " . $e->getMessage();
}
```

Divisibility Check:

Checks for factors from 3 up to the square root of the number.
Only checks odd numbers since even numbers are already handled.

Function to Find and Display Primes in a Range (displayPrimesInRange)

Function Purpose: Finds and displays all prime numbers within a specified range.

Parameters:

\$start: The beginning of the range.
\$end: The end of the range.